

Configurations (.yaml) :: CHEAT SHEET



About

Kaiaulu uses configuration files to store data parameters that are used to run analyses. Instead of having these values in notebooks or scripts, these configuration files let users reuse the same workflows across different projects and analyses.

By splitting the configurations into `project.yaml` (1) and `analysis.yaml` (2) files, it reduces the duplication of data parameters and allows users to mix and match different project and analysis configs.

Project Configuration

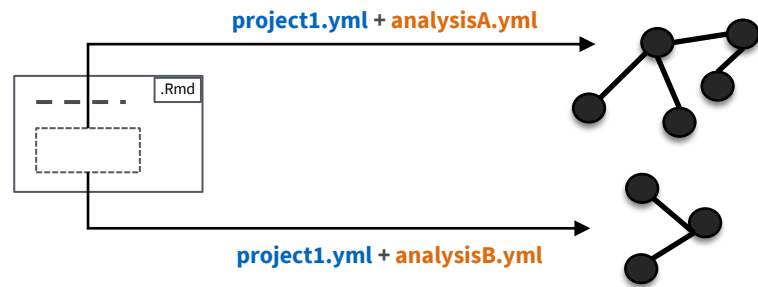
In the configuration, `project.yaml` files (1) define project-related data such as version control, issue trackers, and mailing lists.



These files are specific to a project and remain consistent across different analyses.

Analysis Configuration

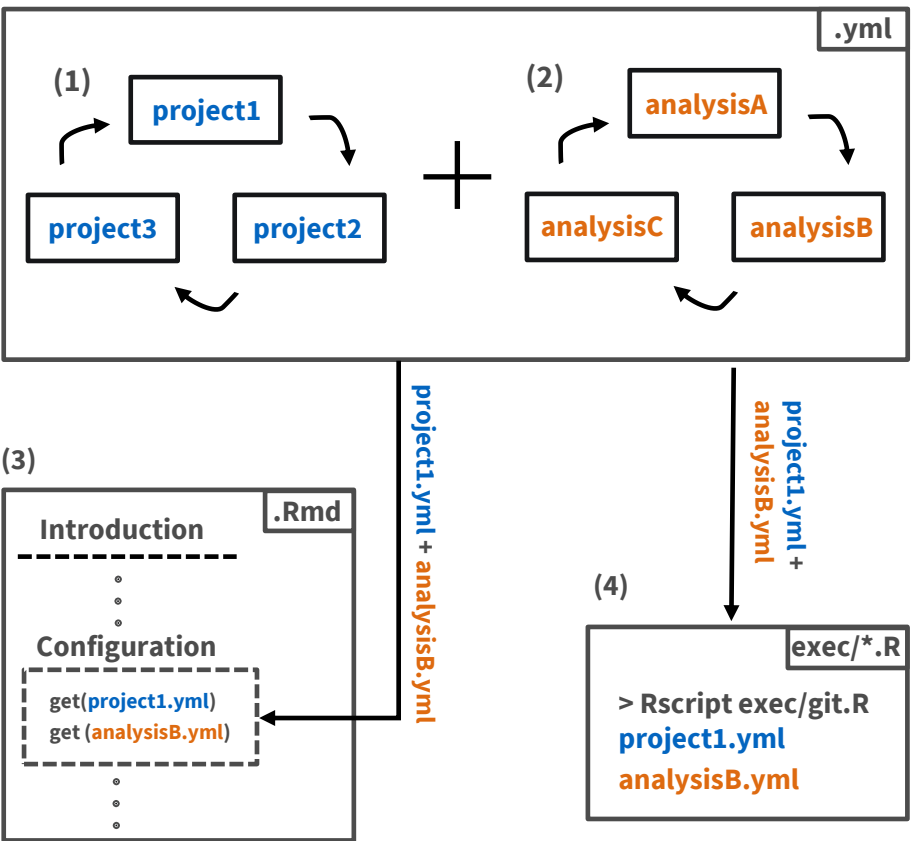
Users can define in the `analysis.yaml` (2) configuration files filters heuristics and tool parameters (e.g. depends, uctags).



These files can be reused across different projects and define how data is analyzed.

Interchangeable Configurations

`Project.yaml` files (1) are combined with any `analysis.yaml` files (2). This allows users to run the same analysis on multiple projects or run different analyses on the same project, reducing duplication of configuration sections. The combined configuration can be used both in notebooks and scripts.



In the notebooks (3), configuration files can be found near the beginning of the file. The notebook parses `project.yaml` (1) and `analysis.yaml` (2) files and gets the values of the parameters with get functions. This allows users to modify parameters in the .yaml files instead of the notebooks.

In the executables (4), both `project.yaml` (1) and `analysis.yaml` (2) files are passed directly as arguments through the command line. The script runs the selected configuration without modification. Although using executables makes `project.yaml` (1) and `analysis.yaml` (2) files fixed during execution, the advantages of using executables versus notebooks are that they can run on server side and do not require knowledge of R.

Config API

Kaiaulu uses get functions to extract values from configuration files. These functions allow notebooks and executables to access parameters in `project.yaml` (1) and `analysis.yaml` (2) files.

Project

The following APIs are used to retrieve parameters from the `project.yaml` configurations (1).

1. `get_git_repo_path()`
Returns the local path of the project's Git repository.
2. `get_jira_issues_path()`
Retrieves the local path to stored JIRA issue data.

Analysis

The following APIs are used to retrieve parameters from the `analysis.yaml` configurations (2).

1. `get_file_extension()`
Returns file types (e.g. c, java) used to filter source files.
2. `get_depends_keep_dependencies()`
Retrieves dependency types specified in analysis configuration.

Temporal Analysis

Temporal parameters in `analysis.yaml` files (2) allow users to analyze a project over time. This allows users to compare data from different timestamps and observe changes and activity in code structure over time.

